

TrES-1b

R-1917

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_130812 · created 2026-08-06T00:14:04+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456516.8522 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.038 +/- 0.02
Transit depth (Rp/Rs)^2	0.15 +/- 0.15 [%]
Orbital Inclination (inc)	89.69 +/- 1.36
Airmass coefficient 1 (a1)	1.016 +/- 0.022
Airmass coefficient 2 (a2)	-0.008 +/- 0.0031
Scatter in the residuals of the lightcurve fit is	2.17 %
Best Comparison Star	#3 - [523, 219]
Optimal Aperture	3.82
Optimal Annulus	8.8
Transit Duration (day)	0.0928 +/- 0.0046

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.038 $\pm$ 0.02 vs 0.1358 (deviation 2.08 σ)
Duration fit vs published	0.0928 d vs 0.1042 d (deviation 1.05 σ)
Mid-time O-C	-2.9 min
Depth SNR	0.8
Residual scatter	2.17 %

Light curve

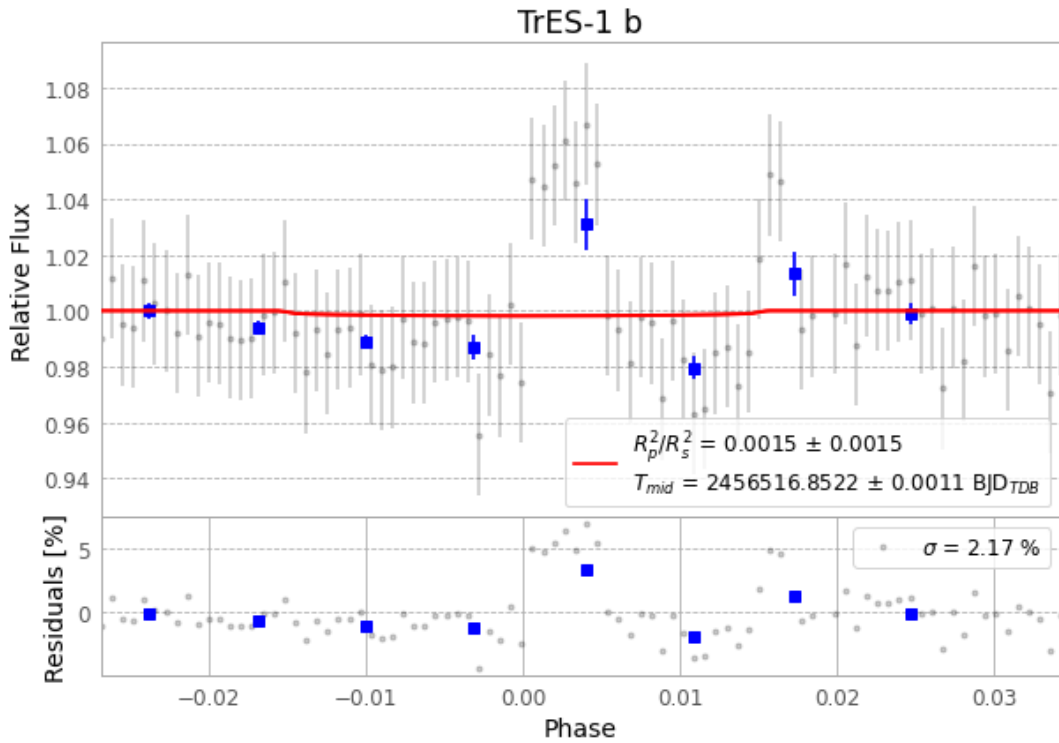


Figure 1 — FinalLightCurve_TrES-1 b_2013-08-11.png (SHA-256 35a6d1377b7dd082...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [421, 315], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5fb22d6f7a8dc706...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

90 FITS frames (59 MB), TRES-1130812062412.FITS → TRES-1130812105112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-130812.log (ab73eeab0681...), FinalLightCurve_TrES-1 b_2013-08-11.png (35a6d1377b7d...), FinalLightCurve_TrES-1 b_2013-08-11.csv (f3138f398695...)

Compute resources

Wall time 177.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-06 00:14Z.